ScholarGraph: Personalized Academic Paper Recommendation System

Dual-Signal AI Architecture Fusing SciBERT Semantic Embeddings with GraphSAGE Citation Topology via LangGraph Agentic Pipeline & Google Gemini LLM

System Abstract & Motivation: Traditional academic discovery engines (e.g., Google Scholar, PubMed) rely heavily on raw keyword frequency (causing vocabulary mismatch) or unweighted citation counts (causing popularity echo-chambers that disadvantage new or cross-domain papers). **ScholarGraph** introduces a dual-signal hybrid framework where **SciBERT (768-d text semantics)** captures conceptual content and **GraphSAGE (128-d inductive topology)** captures academic prestige and citation lineage. The entire lifecycle is orchestrated by a 4-stage **LangGraph StateGraph** agent, enriched with natural-language research briefings generated by **Google Gemini 3.5 Flash-Lite**, and traced in real time using **LangSmith**.

1. System Architecture: Roles, Technologies & Contributions

The system is engineered as an end-to-end pipeline where specialized machine learning and agentic components solve specific stages of discovery:

System Component	Underlying Tech	Primary Responsibility ('Who is Doing What')	System Benefit ('What is it Helping in')
Semantic Intelligence	SciBERT (allenai/scibert_uncased)	Extracts deep representations from titles and abstracts into **768-d dense vectors**.	Resolves terminology mismatch; identifies related papers using distinct vocabularies.
Vector Search Engine	FAISS (IndexFlatIP)	Indexes 56,433 SciBERT vectors; computes exact inner-product (cosine) similarity.	Sub-15ms candidate retrieval from 56k+ literature corpus without exhaustive scans.
Citation Topology GNN	GraphSAGE (PyTorch Geometric)	Aggregates 2-hop neighborhood citations over 103,482 citation edges into **128-d vectors**.	Captures authoritative structural importance; connects papers via community topology.
Agentic Orchestration	LangGraph (StateGraph Engine)	Coordinates 4 state-machine nodes: Intent Analysis, Retrieval, Fusion, Explainability.	Modular, fault-tolerant execution; eliminates rigid monolithic scripts; passes clean state.
AI Senior Advisor	Google Gemini 3.5 Flash-Lite	Analyzes top ranked papers to synthesize overarching themes and recommend anchor paper.	Provides researchers with actionable literature briefings explaining why and what to read first.
Observability & Tracing	LangSmith (Cloud Platform)	Monitors multi-stage latency, tokens, intermediate prompts, and agent state transitions.	Production observability; tracks query trends and verifies agent correctness.
API & Web Dashboard	FastAPI + D3.js + Tailwind CSS	Exposes REST endpoints; renders all-in-one UI with live D3 force graph and library.	Unified, interactive researcher workspace without screen switching or lag.

2. Mathematical Formulation: Dual-Signal Hybrid Fusion

To balance text semantics against structural graph topology, ScholarGraph defines a parameterized fusion score **S_hybrid** between seed paper / query *p* and candidate paper *c*:

Hybrid Scoring Function:

$$S_{hybrid}(p, c) = \alpha \cdot S_{semantic}(p, c) + (1 - \alpha) \cdot S_{graph}(p, c)$$

1. **Semantic Signal:** $S_{semantic}(p, c) = \cos(\mathbf{e}_p^{(SciBERT)}, \mathbf{e}_c^{(SciBERT)}) = (\mathbf{e}_p \cdot \mathbf{e}_c) / (||\mathbf{e}_p|| \cdot ||\mathbf{e}_c||)$
2. **Citation Signal:** $S_{graph}(p, c) = \cos(\mathbf{z}_p^{(GraphSAGE)}, \mathbf{z}_c^{(GraphSAGE)}) = (\mathbf{z}_p \cdot \mathbf{z}_c) / (||\mathbf{z}_p|| \cdot ||\mathbf{z}_c||)$
3. **Dynamic Parameter α ($\alpha \in [0, 1]$):** Default $\alpha = 0.60$ empirically maximizes Mean Reciprocal Rank.

3. End-to-End Query Lifecycle (LangGraph State Machine)

Every recommendation request executes sequentially across 4 distinct agent nodes in LangGraph:

Stage 1: IntentAnalyzerNode (~1ms) — Parses input type (Paper Title vs Topic Query vs Personal Reading List). If a title matches the catalog, it resolves the canonical Paper ID and citations; if topic keywords, it configures free-text vector projection.

Stage 2: SemanticRetrieverNode (~15ms) — Queries the 56,433-vector FAISS index using SciBERT 768-d embeddings to retrieve the top 60 semantic candidate papers, pruning the search space by 99.9%.

Stage 3: GraphSAGEFusionNode (~2ms) — Looks up GraphSAGE 128-d structural vectors for candidate papers, evaluates cosine similarity against seed topology, and computes final hybrid ranking using the α weight.

Stage 4: ExplainabilityAgentNode + Gemini Synthesis (~1.6s) — Formulates multi-perspective explanation metrics (Semantics % vs Graph %) and invokes Google Gemini 3.5 Flash-Lite to produce a scholarly summary and identify the foundational reading anchor.

4. Scientific Benchmark & Evaluation Results

Evaluation strictly follows a **Leakage-Safe Held-Out Protocol**. The 103,482 citation edges were partitioned into 85% train, 5% val, and 10% test. Test citation links were completely excluded from the adjacency matrix during GraphSAGE evaluation.

Recommendation Methodology	MRR (Mean Reciprocal Rank)	Hits@5	Hits@10	Hits@20	ROC-AUC
SciBERT Only (Text Semantics)	0.2334	29.40%	45.10%	58.20%	N/A
GraphSAGE Only (Citation Topology)	0.3891	42.10%	57.30%	68.40%	73.79%
ScholarGraph Hybrid (α = 0.60)	**0.4672 (+100.2%)**	**53.30%**	**65.50%**	**71.00%**	**73.79%**

5. Codebase Structure & Production Implementation

The project codebase is partitioned into active production modules, offline data pipelines, and research evaluation scripts:

backend/app.py: FastAPI web application exposing endpoints for recommendations, fast autocomplete search, D3 subgraphs, and evaluation metrics.

backend/hybrid_recommender.py: Core mathematical engine containing SciBERT tokenizer, FAISS vector search index, and GraphSAGE inference tensors.

backend/agentic_workflow.py: LangGraph StateGraph engine, multi-tier Google Gemini model client with fallback, and LangSmith cloud tracing configuration.

frontend/index.html: High-performance all-in-one dashboard featuring live D3 force citation graphs, library centroid personalization, and benchmarks.

pipeline/: Offline data collection, SciBERT batch embedding generation, FAISS indexing, and PyTorch Geometric GraphSAGE training.

experiments/: Held-out edge evaluation, GPU acceleration benchmarks, similarity diagnosis, and research test harnesses.